How Does Social Behavior Affect Your Password?

Daojing He, Hang Yu, Beibei Zhou, Shanshan Zhu, Min Zhang, Sammy Chan, and Mohsen Guizani

Abstract

A password is an important method of identity authentication, so there are endless research attempts on passwords, but most of them are based on different languages and technologies. As far as we know, no one has studied the influence of one's social behavior (e.g., religious belief) on passwords. Based on this, we first study the influence of religion on the generation of user passwords. Taking Christianity as an example, we find that the passwords of Christians have very strong similarities by analyzing the popular passwords, password structure, letter distribution and length distribution. Furthermore, we find that they contain many words that stem from the Bible, such as Jesus, the Sun, and so on. In order to explore the influence of religious beliefs on password generation, we also select two ordinary social password datasets of non-Christians, and find that they have different characteristics. Finally, we use the classic PCFG attack method to test the anti-attack ability of the Christian password datasets and the ordinary password datasets. Interestingly, we come to the conclusion that the ability of Christian password datasets to resist attacks has a strong similarity, while ordinary password datasets do not have this feature. In particular, by adding Christian information to the rule generated by the ordinary password datasets, we show that it can effectively improve the cracking rate of password attacks.

Introduction

In all aspects of our daily life, we use passwords, such as logging into Internet sites, paying for purchased goods, accessing social accounts, logging into computer systems, and so on. A password is the most primitive way of identity authentication, and it does not provide complete security guarantees. However, even though there are biometric authentication methods such as fingerprint, face or iris, the most commonly used method is still the use of a password. In the digital age and in the mobile Internet era, a password is an irreplaceable verification method. In particular, during the password creation process, the restrictions imposed by various websites and organizations continue to increase, and the selection of a robust password is no longer a simple text string or word. On the other hand, in recent years, the number of hackers has increased exponentially, reminding us to pay attention to select a strong password. For this reason, we need to conduct in-depth password security research and improve the protection of user passwords.

However, the current research on password security mostly focuses on the research of passwords in different languages, and rarely explores the influence of religious beliefs on password creation. Christianity, Buddhism, and Islam are called the three major religions. Among them, Christianity may be the world's largest religion in terms of scale and influence. Christianity originated from Judaism and the faith is centered on Jesus Christ. Its followers are mainly distributed in Europe, America and Oceania, covering 242 countries. At present, the world's population is 7 billion, of which there are 3 billion Christians, accounting for 43 percent of the world 's population [1].

The difficulty of password security research is that passwords are generated by humans and are directly related to human behaviors. Everyone's behavior varies greatly due to internal and external influence. For example, when registering a platform account, some people tend to include the year of birth in the password creation, so their passwords often contain words such as "1995," while some Facebook users tend to include "facebook" in the password. As religious beliefs affect people's behavior, we believe that they are also closely related to the user's password generation. Therefore, it is meaningful to study the impact of religious beliefs on creating passwords.

As early as 2014, Stobert *et al*. [2] mentioned that users often build passwords that include personal related information. They discovered that one participant's password can contain religious phrases such as "God is good." In 2018, Wei *et al*. [3] observed how often passwords were specific to the services for which they were created. They used the religious part extracted from the passwords as the religious theme and found Christian passwords commemorated "jesuschrist" and "john316." However, they ignored the proportion of religious believers in human beings and the influence of religious beliefs on the creation of passwords. In 2019, Yu *et al*. [4] proposed both hierarchical segmentation/optimization algorithms and password prefix/postfix graphs (P3G) construction. They chose a few positive connotative words found in user passwords that were related to faith and religion such as "god," "faith," "jesus." However, we can see that the current password research lacks specialized research on religious passwords, which prompts us to further analyze the characteristics of religious passwords. Therefore, we tend to dig and analyze the religious features in text-based passwords. A good password should have high usability and security. In this

Digital Object Identifier:
10.1109/MNET.101.2000762

Daojing He is with East China Normal University, and also with Harbin Institute of Technology; Hang Yu, Beibei Zhou, and Shanshan Zhu are with East China Normal University; Min Zhang is with Harbin Institute of Technology; Sammy Chan is with City University of Hong Kong; Mohsen Guizani is with Qatar University.

work, we empirically investigate approximately 62 thousands Internet passwords and aim to explore the use of religious belief information in these passwords.

The contribution of this work involves two aspects. On the one hand, to the best of our knowledge, this is the first time to propose analysis of the religious belief information in a password, and study the four password characteristics in the religious social password set, including popular passwords, password structure, letter distribution and length distribution. On the other hand, we compare the ability of religious social passwords and ordinary social passwords to resist guessing attacks, and give some effective suggestions to improve password strength. We improve the cracking rate of password guesstimation on the religious social group by adding religious information to the ordinary social group.

The rest of this article is organized as follows. The next section presents work related to this research topic. We then introduce the password datasets we used and summarize our observations on the characteristics from different groups. Then we study the ability of the selected password datasets to resist attacks and show the influence of religious information in password cracking. Following that, we are committed to improving users' password security awareness by providing some effective suggestions. Finally, we conclude the article.

Related Work

The direction of research on passwords is mainly divided into password data analysis, password guessing attacks, password strength evaluation, and password strengthening. This article mainly studies the influence of religion on user password generation and resistance to attacks, taking Christianity as an example. Therefore, our related work mainly revolves password data analysis and password guessing attacks. In addition, we also highlight some research work on Christianity.

Password Data Analysis

In 2014, Li *et al*. [5] studied the differences between passwords chosen by Chinese and English speaking users. They found that Chinese prefer digits when composing their passwords while English users prefer letters. In 2018, AlSabah *et al.* [6] analyzed a meta-data rich data leak from a Middle Eastern bank with a demographically-diverse user base. They illustrated the differences that existed in how users from different cultural/linguistic backgrounds create passwords. In 2019, Wang *et al.* [7] performed an extensive, empirical analysis of 73.1 million real-world Chinese web passwords in comparison with 33.2 million English counterparts. Their cracking results revealed the bifacial-security nature of Chinese passwords. In 2020, Kavrestad *et al*. [8] discussed how users who actively encrypt data constructed their passwords to benefit the forensics community. Their results suggested that those users were more likely to use keyboard patterns as passwords than other users. According to the above research, user language plays a significant role in password creation. However, in [3], it is mentioned that religious themes also influenced password creation. Therefore, ignoring religious factors may prevent us from discovering password characteristics. This motivated us to investigate how passwords are chosen when influenced by the Christian religion.

Password Guessing Attack

First, Weir *et al*. [9] proposed a PCFG, which is an algorithm based on probabilistic context-free grammar. They generated word-mangling rules based on the grammar and used them to guess passwords. In 2014, Egwali *et al*. [10] presented an empirical investigation into users' regular passwords (RPs) and mnemonic passwords (MPs). They found that a total of 121 users' MPs were in the MPs dictionary relating to respondents' names, personal information, religious and family background. In 2017, Liu *et al*. [11] presented the first approach for measuring the security of gesture passwords with guessing attacks that model real-world attacker behavior. In 2018, Wang *et al*. [12] performed a large-scale empirical analysis of password reuse and modification patterns and developed a new training-based guessing algorithm. In 2019, Mori *et al*. [13] demonstrated that the knowledge of the linguistic background of targeted users can be exploited to increase the speed of the password guessing process.

Christian Studies

In 2002, Placher *et al*. [14] mentioned Jesus Christ is also seen as universal, being able to relate to diverse cultures and time periods, transcending languages, gender, and life experiences. In 2020, Karuvelil [15] mentioned that diversity of religions is explained in terms of the institutionalization of diverse experiences. Both articles talk about the influence of Christian culture on many aspects of people's lives. Inspired by this, we consider adding Christian vocabulary words to the training rule in order to achieve better experimental results, which can be found below.

Despite the fact that many aspects have been studied, until now little work has concentrated on password security research based on religion. The above research attempts encouraged us to further study the features of passwords that are created by religious individuals.

Password Analysis

To better describe the characteristics of Christian passwords, two password datasets of different web services for Christians and two social networking sites are used in our study. At the same time, we categorize these four datasets into two groups, one is the Christian social group (according to the users of websites), and the other is the ordinary social group (according to the type of website services). The following are four aspects of our analysis of passwords.

Popular Passwords: These passwords are the kind of passwords that many people like to use, and they are also the worst passwords on the Internet. These passwords are all high-risk passwords and are easy to crack.

Password Structure: We take five types of password characters into consideration, including lowercase letters (LL) type, uppercase letters (UL) type, mixed uppercase and lowercase letters (ML) type, digits (D) type, and special characters (S) type. We study the proportion of password compositions in different types of groups, such as D, LL, ML+D, UL+D, LL+D.

Letter Distribution: In the case of ignoring the letter case, there are 26 letters in the English language. We analyzed their frequency distribution in each password dataset.

Length Distribution: The frequency distribution of password length in password datasets.

Popular Passwords

In this section, we investigate the top 10 most popular passwords in the datasets of the Christian social group and ordinary social group. As can be seen from Table 1, the most popular password is very weak, which is "123456." In the Christian social group, it is clear that the passwords in both websites contain "jesus" and "christ." The Bible records that Jesus Christ is the Son of the Most High God Jehovah and the person of the Holy Son in the Trinity. More passwords related to the Christian religion can also be found in these two datasets. In Faithwriter, we often find "heaven" which represents the place where the Most High God lives. In addition, the password "john316" appears very frequently. John is one of the twelve apostles of Jesus Christ. It is said that he wrote the "Gospel of John," which describes the main words and deeds of Jesus. In Singles, the password "princess" has a high frequency. Meanwhile, it cannot be ignored that the passwords from Faithwriter are similar to the website name, such as "writer" and "faithwriters," which is significantly different from Singles, which contains the emotion of love.

In the ordinary social group, the passwords "123456" and "password1" are the top weak passwords of the two websites, respectively. Obviously, the keyboard scheme and emotional password appear in it, such as "qwerty" and "iloveyou1." In Facebook, we find many passwords related to people's names, such as "lincoln" and "celeste." As we all know, Abraham Lincoln is the 16th President of the United States and celeste is used to describe a person who is very smart and talented. In Myspace, "monkey1" and "football1" belong to animals and sports, respectively. An interesting finding is that the name of the website was used as a password.

Top 10	Christian social group		Ordinary social group	
	Faithwriter	Singles	Facebook	Myspace
1	123456	123456	123456	password1
2	writer	jesus	123456789	abc123
3	jesus1	password	com	fuckyou
4	christ	12345678	lincoln	monkey1
5	blessed	christ	123654789	iloveyou1
6	john316	love	qwerty	myspace1
7	jesuschrist	princess	newron	fuckyou1
8	password	jesus1	12345678	number1
9	heaven	sunshine	farmville	football1
10	faithwriters	1234567	celeste.0206	nicole1

TABLE 1. Popular passwords of two groups.

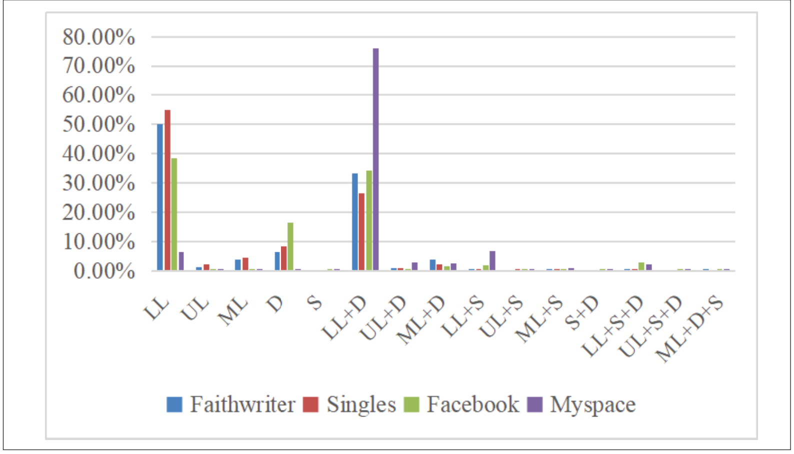


FIGURE 1. Proportion of password structure in two groups.

Password Structure

As an essential part of a password, characters are worthy of our detailed study. The password structures have a huge impact on password strength. Moreover, password structure is not only influenced by users' own behaviors, but also affected by the categories of groups.

In Fig. 1, it is obvious that the structure LL has the highest proportion in the Christian social group. The most likely reason is that Christians are mainly distributed in Europe and America, and most of them are English users. As mentioned in [5], English users prefer letters when composing their passwords. Besides, we find that special characters (S) are basically not used, indicating that they are not welcomed by Christian users. Other than that, we can see that the proportion of any structure of Faithwriter is very small compared to Singles.

In the Ordinary social group, Facebook and Myspace show an obvious difference. In Facebook, LL accounted for the largest proportion, but LL+D accounted for the largest proportion in Myspace. What's more surprising is that D's proportion in Facebook has reached 16.37 percent, while it is almost 0 in Myspace.

Letter Distribution

In our intuition, English words and English passwords have a lot of difference. Usually, in some way, we can say that the password creation is influenced by users' mother tongue. Thus, the letter frequency of the English words may be different from that of English passwords. In order to explore their differences, we ignore the issue of letter case when calculating the distribution of letters. The analysis results are shown in Fig. 2. As we can see, the letter distribution in these four datasets shows a similarity. This phenomenon reveals that sometimes the letter frequency distribution of passwords is not affected by the type of website service, and may be related to the user's language. Apparently, the letters "a", "e", "i", "o", "r" appear frequently in these four password datasets. According to the popular password analysis result, "jesus1" and "christ" appear frequently. This may explain the ordering of these alphabetic frequencies.

When it comes to the letter distribution of the Christian social group, their differences are mainly reflected in several letters, such as "r" and "o". For example, the letter "r" in Faithwriter appears more frequently than in Singles. It is probably because its top 10 popular passwords contain the letter "r", such as "writer," "christ" and "faithwriters." Conversely, the letter "o" in Faithwriter has less frequency than in Singles. Similarly, it is because its top 10 popular words contain the letter "o", such as "love" and "password."

In the ordinary social group, it is apparent that letters "b" and "y" in Myspace have higher frequencies than in Facebook. It is probably because its top ten popular words contain the letters "b" and "y", like "abc123"and "monkey1." We can see that the frequency of the letter "n" in Facebook is higher than in Myspace. By the same token, it can be explained that its top 10 popular words contain the letter "n", such as "lincoln" and "newron." More importantly, after ignoring the subtle differences between the letter frequency distribution, we can find that the curve trends of these four English datasets are very similar.

Length Distribution

Length also is an essential property of a password. It cannot be denied that the password length has a huge impact on password strength.

It can be seen from Fig. 3 that about 70 percent of Christians have passwords lengths between 7 and 9. Most importantly, we can see that in Singles, passwords with a length of 8 are the most created, while passwords with a length of 6 are the most popular in Faithwriter. Other than that, we can see that almost nobody has a password length longer than 13. This means the probability of password cracking will significantly improve.

In the ordinary social group, we can see that more than 85 percent of users have password lengths between 6 and 13. Users of both websites are most inclined to create passwords of length 8. In addition, there are almost no passwords longer than 13 in Myspace, but passwords longer than 13 in Facebook account for 9 percent of the total.

PCFG-Based Attacks

Based on the analysis in the previous section, we can see that users of two password datasets have many differences in characteristics and preferences when generating passwords. At the same time, they are surprisingly similar in letter frequency distribution. Therefore, we employed an advanced password attacking model (PCFG [9]) to evaluate the strength of passwords of Faithwriter and Singles. The PCFG algorithm and Markov algorithm are the current mainstream password guessing algorithms. The PCFG algorithm's fitting effect on real password data has been proven to be better than the Markov algorithm. In the PCFG attacking algorithm, the password is divided into a letter segment L, a digit segment D, and a special character segment S. The segments are independent of each other. For example, the password "tian28@!@" is divided into $L_4D_2S_3$, which is called the base structure. Then, through statistical knowledge, the probabilities of all base structures and segments in the training set are calculated. According to this method, the probabilities of all passwords are obtained. For example, the formula for the probability calculation of the password "tian28@!@" is as follows: $P(tian28@!@) = P(L_4D_2S_3) \times P(L_4 \rightarrow tian) \times P(D_2 \rightarrow 28) \times P(S_3 \rightarrow @!@)$. For comparison, we added an evaluation of the Facebook and Myspace datasets. To detect the role of Christian faith against resisting password guessing attacks, we conduct an advanced experiment.

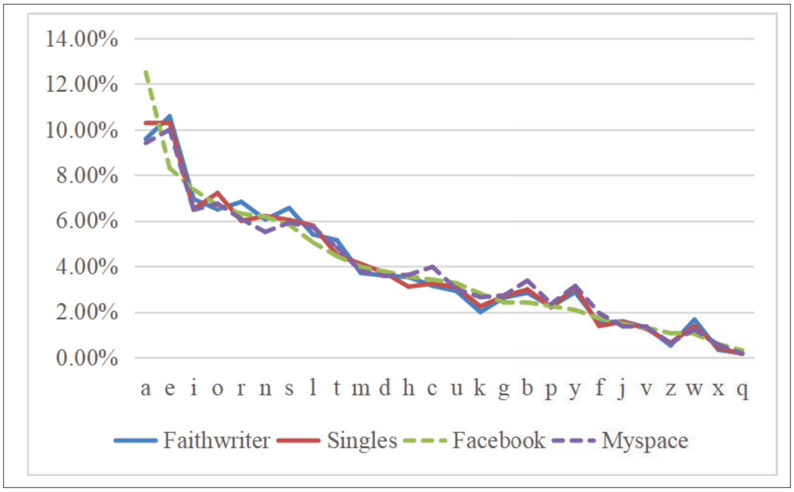


FIGURE 2. Letter distribution in two groups.

Training Dataset Selection

The PCFG-attack algorithm that we chose needs a training dataset. In addition, the quality of the training dataset will affect the rationality of the results. To ensure fairness, we randomly selected half of the passwords from each of the two password datasets, and combined them as the training dataset. There are 10,791 passwords in the training dataset. In the same way, we determined the training set of Facebook and Myspace datasets. There are 19,792 passwords in the training dataset.

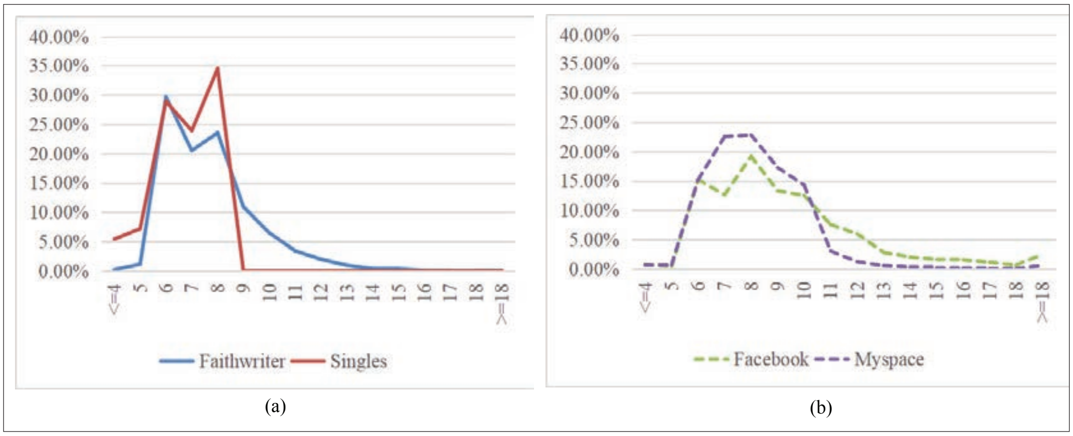


FIGURE 3. Length distribution of two groups: a) length distribution of Faithwriter and Singles; b) length distribution of Facebook and Myspace.

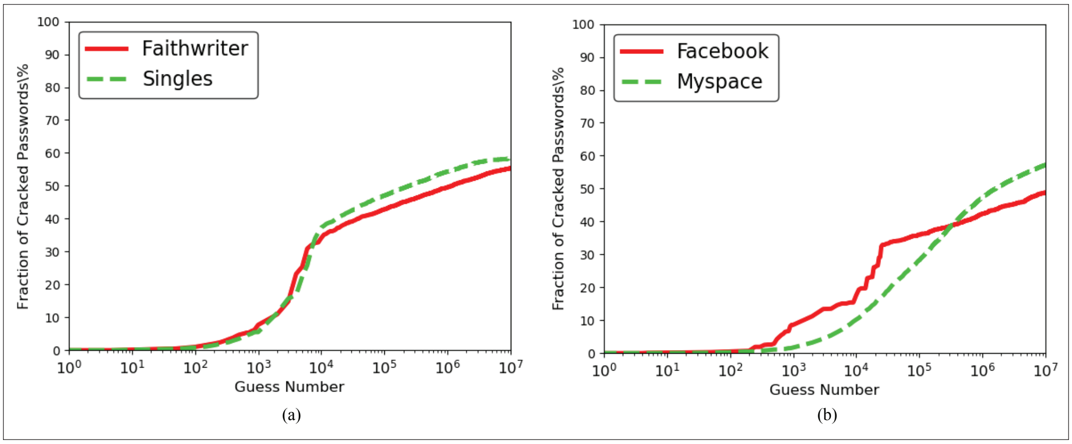


FIGURE 4. PCFG attack on two groups: a) PCFG attack on Faithwriter and Singles; b) PCFG attack on Facebook and Myspace.

Experimental Results

Applying PCFG on the four datasets, the cracking results are shown in Figs. 4a and 4b. We should note that in this experiment, the number of our password guesses did not exceed 10^7.

In Fig. 4a, we have the following findings. When the number of guesses does not exceed 10^4, we can see that the password strength of the two password datasets is very similar (we can see that the two curves in the figure almost overlap). In addition, the success rate of the Singles is slightly lower than that of the Faithwriter. When the guess number exceeds 10^4, it shows an opposite result. Single's cracking rate is 1.9 percent higher than Faithwriter's cracking rate. According to the length distribution analysis, the longest password length in the Singles is 8, so when the number of password guesses becomes larger, the passwords of the Singles appear weaker than that of the Faithwriter.

From Fig. 4b, we can see that the findings are similar to those in Fig. 4a. For both social networking sites, the password strengths of these two datasets only change when the number of guesses reaches about 300,000. When the number of guesses is lower than this threshold, the Myspace passwords have a higher strength, but the Facebook passwords become more secure after it exceeded the threshold.

Advanced Experimental Results

As we know, the core of PCFG is the password rules generated after training. For this reason, we modified the rule generated by the training on Facebook and Myspace datasets. We added 2000 vocabulary words related to Christianity to the rule. In the rules, these vocabulary words are allocated with the highest probabilities. There are two rule sets, one is the unmodified rule, and the other is the modified rule. We use them to attack the Christian social group respectively. The results are shown in Fig. 5.

We can see that when Christian information is not included, the cracking rate is no more than 35 percent, as shown in Fig. 5a. After adding the related information, the cracking rate exceeds 40 percent, as shown in Fig. 5b. By comparison, we found that the cracking rate has increased by 6 percent to 8 percent. Given that an attacker has ordinary social networking datasets, he can improve the success rate of cracking Christian password datasets by including Christian information.

Summary

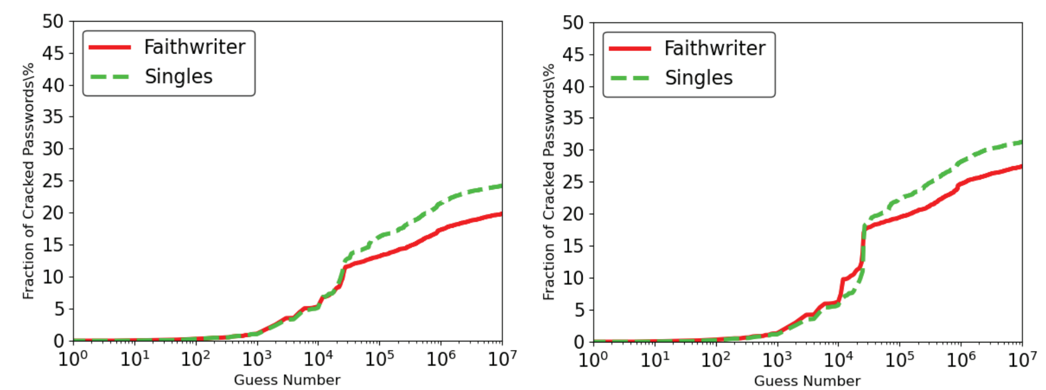
As we can see, the cracking rate increases as the number of password guesses increases. In Fig. 4a, an interesting phenomenon cannot be ignored. Although the two websites provide very different services, that is, writing service and match making service, we found that the cracking rate of the two is not much different (the number of guesses is less than 10^7). This is probably related to the fact that the users of these two websites are both Christians. However, in Fig. 4b, we found a different result. For both websites that provide social services, the difference in cracking rates between the two is more obvious. According to this phenomenon, we can see that the password strength of the website is not only related to the type of website service, but also to the religious belief of website users. In Fig. 5, we find that the inclusion of Christian information can significantly increase the cracking rate.

Some Suggestions

We suggest that Christians use "jesus" and "christ" as little as possible when constructing their own passwords, which will greatly improve the security of passwords. Using a longer password will make the password more robust. According to our analysis of the password structure, adding special characters when creating a password will also greatly reduce the risk of the user's password being cracked. When analyzing passwords according to the types of website services, we can also take into account the user's behavior. In this article, we consider religious belief and also show its effect on password guessing. What needs to be added is that attackers may build a better dictionary according to the user's behavior, thereby reducing the size of the dictionary and increasing the password cracking rate. For example, a series of factors that affect the user's password creation behavior, such as their belief, hobby or nationality, can be exploited. Therefore, we hereby urge users not to disclose too much personal information when setting passwords.

Conclusion

With the increase of password length and complexity of the composition, it is critical to understand a password's characteristics to ensure its security.



(a) (a)

FIGURE 5. PCFG attack on Faithwriter and Singles: a) guessing on original rule; b) guessing on modified rule.

In this article, we carried out a large-scale empirical analysis and made some recommendations. We analyzed these passwords from four aspects, including popular passwords, password structure, letter distribution and length distribution. We can conclude that Christians have very similar password settings. In order to compare the anti-attack capabilities of Christian social password datasets and ordinary social password datasets, we applied the PCFG algorithm to conduct attack experiments. The results suggested that there is a close relationship between the users' religious beliefs and the ability of passwords to resist attacks. These findings present helpful insights for users to select more secure passwords. Then, by considering a variety of factors that affect human behavior, researchers may improve the ability of passwords to resist attacks and the level of Internet security. In the future, we will carry out further theoretical research on the influence of religious beliefs on password generation.

Acknowledgment

This research is supported by the National Natural Science Foundation of China (Grants: U1936120, U1636216); the National Key R&D Program of China (2017YFB0802805 and 2017YFB0801701); and the Basic Research Program of the State Grid Shanghai Municipal Electric Power Company (52094019007F). Daojing He is the corresponding author of this article.

References

[1] Wikipedia, "Christians," https://en.wikipedia.org/wiki/Christians, 2021.
[2] E. Stobert and R. Biddle, "The Password Life Cycle: User Behaviour in Managing Passwords," *Proc. 10th Symposium on Usable Privacy and Security (SOUPS)*, Menlo Park, July 9–11, 2014, pp. 243–55.
[3] M. Wei *et al.*, "The Password Doesn't Fall Far: How Service Influences Password Choice," *Who Are You (WAY)*, Aug. 2018.
[4] X. Yu and Q. Liao, "Understanding User Passwords Through Password Prefix and Postfix (P3) Graph Analysis and Visualization," *Int'l. J. Inf. Sec.*, vol. 18, no. 5, Sept. 2019, pp. 647–63.
[5] Z. Li *et al.*, "A Large-Scale Empirical Analysis of Chinese Web Passwords," *Proc. 23rd USENIX Security Symposium*, San Diego, Aug. 20–22, 2014, pp. 559–74.
[6] M. AlSabah *et al.*, "Your Culture Is in Your Password: An Analysis of a Demographically-Diverse Password Dataset," *Computers & Security*, vol. 77, no. 23, Aug. 2018, pp. 427–41.
[7] D. Wang *et al.*, "Birthday, Name and Bifacialsecurity: Understanding Passwords of Chinese Web Users," *Proc. 28th USENIX Security Symposium*, Santa Clara, Aug. 14–16, 2019, pp. 1537–55.
[8] J. Kavrestad *et al.*, "Analyzing the Usage of Character Groups and Keyboard Patterns in Password Creation," *Inf. Comput. Secur.*, vol. 28, no. 3, Aug. 2020, pp. 347–58.
[9] M. Weir *et al.*, "Password Cracking Using Probabilistic Context-Free Grammars," *Proc. 30th IEEE Symposium on Security and Privacy (IEEE S&P)*, Washington, May 3-5, 2009, pp. 391–405.
[10] A. Egwali and E. Onibere, "Users Mnemonic Passwords: An Empirical Analysis," *J. Computer Science and Its Application*, vol. 21, no. 1, May. 2014, pp. 10–18.
[11] C. Liu *et al.*, "Guessing Attacks on User-Generated Gesture Passwords," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 1, no. 1, Mar. 2017, pp. 1–24.
[12] C. Wang *et al.*, "The Next Domino to Fall: Empirical Analysis of User Passwords Across Online Services," *Proc. 8th ACM Conf. Data and Application Security and Privacy (CODASPY)*, Tempe, Mar. 19–21, 2018, pp. 196–203.
[13] K. Mori *et al.*, "Comparative Analysis of Three Language Spheres: Are Linguistic and Cultural Differences Reflected in Password Selection Habits?" *Proc. 4th IEEE European Symposium on Security and Privacy (EuroS&P)*, Stockholm, June 17–19, 2019, pp. 159–71.
[14] W. Placher, "Jesus the Savior: The Meaning of Jesus Christ for Christian Faith," *Concordia J.*, vol. 28, no. 1, Jan. 2002, pp. 36–37.
[15] G. Karuvelil, "Religious Diversity, Christian Faith, and Truth," *Faith, Reason, and Culture: An Essay in Fundamental Theology*, pp. 317–47. July 2020.

Biographies

Daojing He [S'07, M'13] received the B.Eng. (2007) and M. Eng. (2009) degrees from Harbin Institute of Technology, China, and the Ph.D. degree (2012) from Zhejiang University, China. He is currently a professor at East China Normal University, P.R. China. His research interests include network and systems security.

Hang Yu is currently a master student in the School of Software Engineering, East China Normal University, P.R. China.

Beibei Zhou is currently a Ph.D. student in the School of Software Engineering, East China Normal University, P.R. China.

Shanshan Zhu is currently working at East China Normal University, P.R. China. She received the bachelor degree from Huazhong University of science and technology, China, and the Master degree from Zhejiang University, China, in 2007 and 2010, respectively,

Min Zhang received the bachelor and Ph.D. degrees in computer science from the Harbin Institute of Technology, Harbin, China, in 1991 and 1997, respectively. He is currently a distinguished professor at Harbin Institute of Technology, Shenzhen, P.R. China. His current research interests include natural language processing and artificial intelligence.

Sammy Chan [S'87-M'89) received a Ph.D. degree in communication engineering from the Royal Melbourne Institute of Technology, Australia, in 1995. Since December 1994 he has been with the Department of Electrical Engineering, City University of Hong Kong, where he is currently an associate professor.

Mohsen Guizani [S'85, M'89, SM'99, F'09] received the B.S. (with distinction) and M.S. degrees in electrical engineering, and the M.S. and Ph.D. degrees in computer engineering from Syracuse University, Syracuse, NY, USA, in 1984, 1986, 1987, and 1990, respectively. He is currently a professor at the CSE Department at Qatar University, Qatar.